ENGR 111
Introduction to Engineering
Class #1

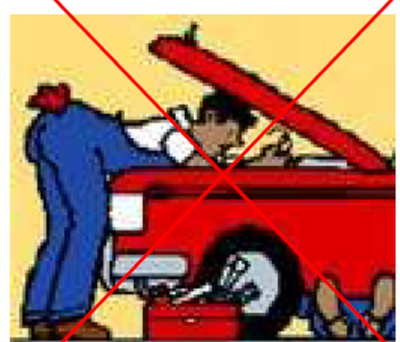
What is engineering?

Who is NOT an engineer?

Plumber

Electrician

Carpenter



Auto Mechanic

Related question: Where does engineering end and skilled technical work begin?

Who is NOT an engineer?

PC Technician

Building Supervisor/Handyman

Welder

Machinist

200 years of trying to define engineering

1799 — Count Rumford (Physicist & inventor; pioneer in heat science)
"The application of science to the common purpose of life."

1887 — A. M. Wellington (Civil/railroad engineer; pioneer of engineering economics)
" […] It is the art of doing that well with one dollar which any bungler can do with two […]."

1954 — M. P. O'Brien (Civil engineer; pioneer of coastal engineering)
"[It] is the design of structures, machines, circuits, or processes […] and the analysis and prediction of their performance and costs […]"

1962 — Gordon S. Brown, MIT (Electrical engineer; pioneer in feedback and control systems)
"[It] is practicing the art of the organized forcing of technological change […] Engineers operate at the interface between science and society […].."

SCIENCE → ECONOMICS → DESIGN & ANALYSIS → TECHNOLOGICAL CHANGE

What do Engineers do?
(generic modern definition)

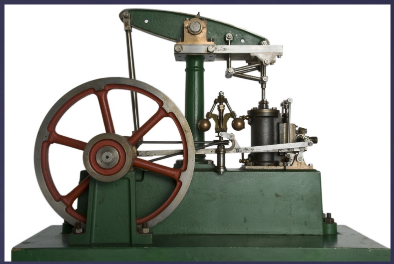
Engineers apply scientific principles to design, build, and improve systems, structures, and processes that solve real-world problems.

Their work is focused on practicality, efficiency, and functionality, often balancing technical constraints with economic, environmental, and regulatory considerations.

The long arc of engineering

Stone tool → Wheel → Steam engine → Electric power → Electronics → Smartphone

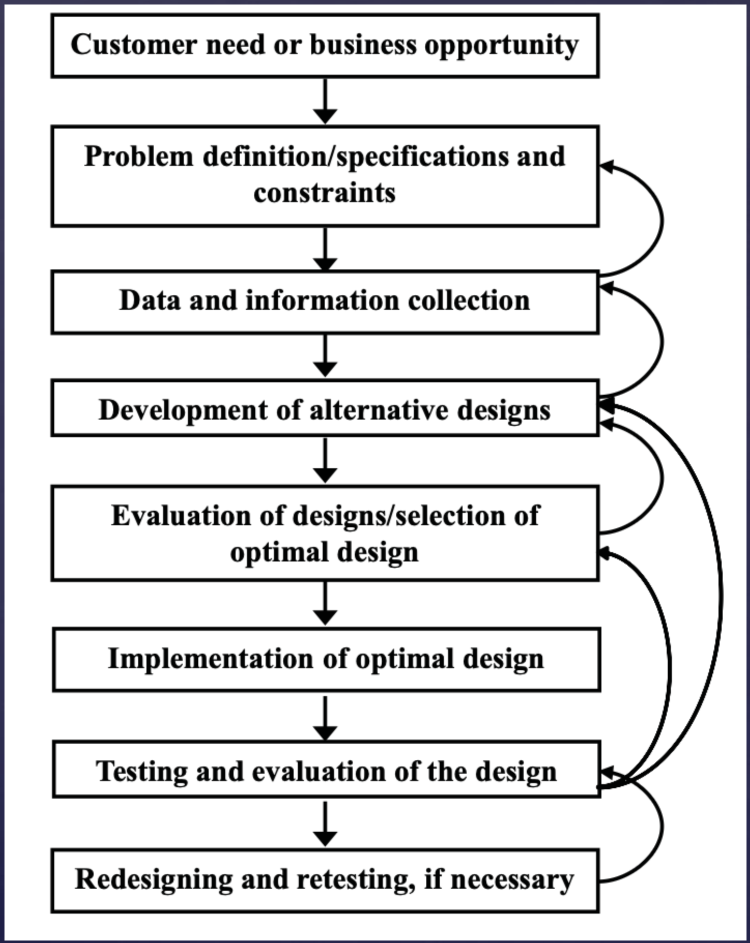





Engineering existed as a human activity long before “engineer” became a profession.

Common thread: People started with problems, needs, ideas, accidents, observations, or just tinkering. They tried things, learned what happened, and improved what came before.

The engineering design process



- Process that ties modern engineering together.
- Engineers may work on the whole process or specialize in only one part of it.
- We will study this process in more detail in a later class.

Need → Define → Explore → Design → Build → Test → Improve

The National Academy of Engineering's top 20 achievements of the 20th century

20. High performance materials
19. Nuclear technologies
18. Laser and fiber optics
17. Petroleum and gas technologies
16. Health technologies
15. Household appliances
14. Imaging technologies
13. Internet
12. Space exploration
11. Interstate highways

The National Academy of Engineering's top 20 achievements of the 20th century (continued)

10. Air-conditioning and refrigeration
9. Telephones
8. Computers
7. Agricultural mechanization
6. Radio and television
5. Electronics
4. Safe and abundant water
3. Airplanes
2. Automobiles
1. Electrification

Engineering in the 21st century

- What engineering achievements of the 21st century already belong on a top 20 list?
- What do you think will define the rest of the century?

Why do you want to be an engineer?

1. What matters most to you in a career?
Compensation • Work life balance • Impact • Challenge • Career growth • Security • Flexibility • Culture

2. What matters most to you in a career?